



EXAM PAPERS PRACTICE

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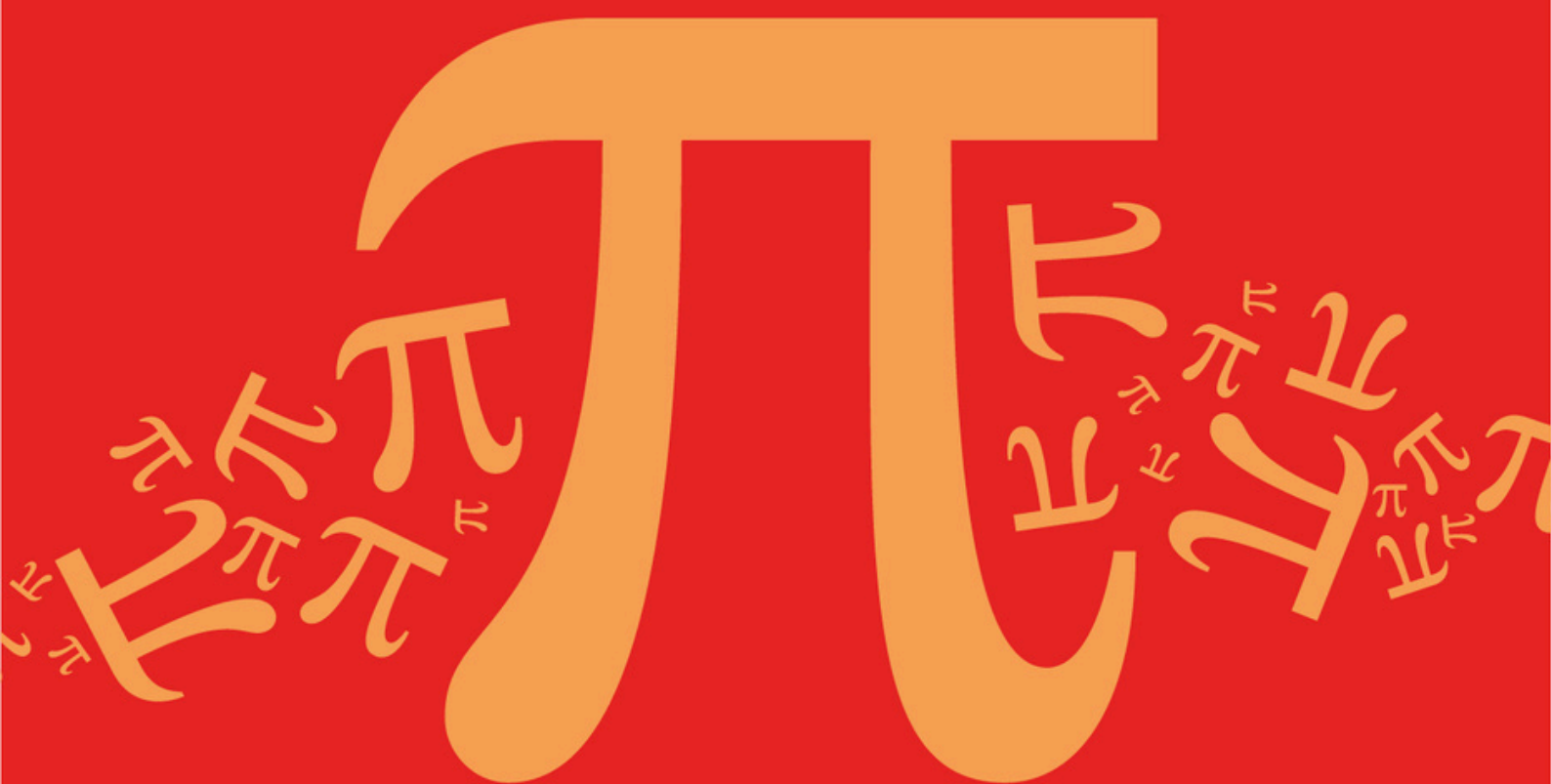
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Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D4: Method of Differences - Use of Partial Fractions

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

D: Further Algebra and Functions

Sub topic

D4: Method of Differences - Use of Partial Fractions

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Given that

$$\frac{2}{(r+1)(r+2)(r+3)} = \frac{A}{(r+1)(r+2)} + \frac{B}{(r+2)(r+3)}$$

find the values of the integers A and B

(2)

- (b) Use the method of differences to show clearly that

$$\sum_{r=9}^{97} \frac{1}{(r+1)(r+2)(r+3)} = \frac{89}{19800}$$

(4)

(Total 6 marks)

Q2.

- (a) Express $\frac{1}{r(r+2)}$ in partial fractions.

(3)

- (b) Use the method of differences to find

$$\sum_{r=1}^{48} \frac{1}{r(r+2)}$$

giving your answer as a rational number.

(5)

(Total 8 marks)

Q3.

- (a) Given that

$$\frac{1}{4r^2 - 1} = \frac{A}{2r - 1} + \frac{B}{2r + 1}$$

find the values of A and B .

(2)

- (b) Use the method of differences to show that

$$\sum_{r=1}^n \frac{1}{4r^2 - 1} = \frac{n}{2n+1}$$

(3)

- (c) Find the least value of n for which $\sum_{r=1}^n \frac{1}{4r^2 - 1}$ differs from 0.5 by less than 0.001.

(3)

(Total 8 marks)

Q4.

- (a) Given that

$$\frac{1}{r(r+1)(r+2)} = \frac{A}{r(r+1)} + \frac{B}{(r+1)(r+2)}$$

show that $A = \frac{1}{2}$ and find the value of B .

(3)

- (b) Use the method of differences to find

$$\sum_{r=10}^{98} \frac{1}{r(r+1)(r+2)}$$

giving your answer as a rational number.

(4)

(Total 7 marks)

Q5.

- (a) Given that

$$\frac{r^2 + r - 1}{r(r+1)} = A + B\left(\frac{1}{r} - \frac{1}{r+1}\right)$$

find the values of A and B .

(3)

- (b) Hence find the value of

$$\sum_{r=1}^{99} \frac{r^2 + r - 1}{r(r+1)}$$

(4)

(Total 7 marks)